

2082

Bachelor of Science in Computer Science and Information Technology

Course Title: **Cryptography**

Code No: **CSC327** Semester: **5th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Describe the Fiestel Cipher structure. Given the key {28,7F,,15, 16, 28, AE, D2, A6, AB, F7, 97,66,01,02,03,04}, compute the first 4 byte of next key after first iteration, using the following S-Box in AES.0123456789ABCDEF0637C777BF26B6FC53001672BFED7AB76
2. Illustrate the man in middle attack in Diffie - Hellman key exchange protocol. Assume the prime number be 19 and 10 as its primitive root. Select 5 as private key and 4 as random integer. Find the cipher text of M = 2 using Elgamal crypto system.
3. List the properties of hash function. Describe the algorithm for SHA -1 .

Section B

Attempt any Eight questions[8×5=40]

4. Describe any three types of malicious logic.
5. Given the key "HELLOWORLD", encrypt the plaintext "TURINGTEST" using Play fair cipher.
6. Explain any two modes of block cipher encryption.
7. State Fermat's theorem with example. What is the implication of discrete logarithm?
8. Describe the working mechanism of digital signature algorithm.
9. List and explain the types of firewall.
10. Decrypt the ciphertext "HI" using Hill Cipher where the key is5341.
11. What is digital certificate? Discuss the certificate life cycle.
12. Define authentication system. Discuss about challenge response system.